WEEK 3 · LESSON 16 OF 24 · HANDOUT

Probability: Measuring Uncertainty

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=OyddY7DlV58 — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 2: Uncertainty (probability and Bayes) — https://www.youtube.com/watch?v=D8RRq3TbtHU

🎓 **Go deeper:** freeCodeCamp: Statistics — a full university course — https://www.youtube.com/watch?v=xxpc-HPKN28

Your goal in this lesson: Calculate basic probabilities, use the complement, "and" and "or" rules, understand independence and conditional probability, and check answers by simulation.

AI lives with uncertainty

A spam filter does not *know* an email is spam; it says *"97% likely"*. A weather model says *"70% chance of rain in Musanze tomorrow"*. **Probability** is the language AI uses to express how sure it is.

The basics

Probability is a number from **0** (impossible) to **1** (certain). For equally likely outcomes:

> *P(event) = number of favourable outcomes ÷ total number of outcomes*

- Rolling a 6 with one die: P = 1/6 ≈ 0.167.
- Picking a girl from a class of 18 girls and 22 boys: P = 18/40 = 0.45.

Three rules

1. Complement — "not": P(not A) = 1 − P(A). If P(rain) = 0.3, then P(no rain) = 0.7.

2. "And" for independent events: P(A and B) = P(A) × P(B). Two dice both show 6: 1/6 × 1/6 = 1/36. Events are **independent** when one does not change the chance of the other.

3. "Or": P(A or B) = P(A) + P(B) − P(A and B). We subtract the overlap so it is not counted twice. If A and B cannot happen together, the overlap is 0.

Conditional probability: "given that"

P(A | B) means *"the probability of A, given that B happened"*:

> *P(A | B) = P(A and B) ÷ P(B)*

Example — 1,000 mobile-money transactions:

	Flagged by system	Not flagged	Total
Fraud	18	2	20
Normal	40	940	980
Total	58	942	1,000

- P(fraud) = 20/1000 = 0.02
- P(fraud | flagged) = 18/58 ≈ 0.31
- P(flagged | fraud) = 18/20 = 0.90

Notice: **P(fraud | flagged) is not the same as P(flagged | fraud).** Mixing them up is one of the most common mistakes in data work — and in news reports. You met these numbers before as **precision** (0.31) and **recall** (0.90).

Check by simulation

When maths feels tricky, simulate thousands of random trials:

```
import numpy as np
rng = np.random.default_rng(0)

die1 = rng.integers(1, 7, 100000)
die2 = rng.integers(1, 7, 100000)
print(((die1 == 6) & (die2 == 6)).mean())   # close to 1/36 = 0.0278
print(((die1 + die2) == 7).mean())          # close to 6/36 = 0.1667
```

The share of trials where the event happens gets closer to the true probability as the number of trials grows — the **law of large numbers**.

Expected value

The **expected value** is the long-run average outcome:

> *E = Σ (value × probability)*

A raffle ticket costs 1,000 RWF. There is a 1 in 500 chance to win 200,000 RWF. Expected winnings = 200,000 × 1/500 = 400 RWF — less than the 1,000 RWF price, so on average you lose 600 RWF per ticket.

Key terms

- **Probability** — a number from 0 to 1 measuring how likely an event is.
- **Complement** — the event "not A": 1 − P(A).
- **Independent events** — one does not affect the other.
- **Conditional probability** — P(A | B), the chance of A given B.
- **Law of large numbers** — with many trials, the observed share approaches the true probability.
- **Expected value** — the long-run average result.

Remember this: Probability measures uncertainty from 0 to 1. Learn the complement, "and" and "or" rules, never confuse P(A | B) with P(B | A), and check your answers with simulation.

Name: _____ Date: _____

Activity

Task 1

A bag has 5 red, 3 blue and 2 green beads. What is P(red)? P(not green)?

Task 2

Simulate drawing one bead 100,000 times with `rng.choice(["red"]*5 + ["blue"]*3 + ["green"]*2, 100000)` and compare with your answers.

Task 3

From the fraud table, calculate P(flagged | normal). What does it mean for customers?

Task 4

Compute the expected value of a game: pay 500 RWF, roll a die, win 2,000 RWF on a 6. Should you play?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. If P(rain) = 0.3, what is P(no rain)?
   - A. 0.3
   - B. 0
   - C. 1.3
   - D. 0.7
2. What is the probability that two fair dice both show 6?
   - A. 1/36
   - B. 1/6
   - C. 2/6
   - D. 1/12
3. In the fraud table, P(fraud | flagged) = 18/58. What does it mean?
   - A. Of all fraud, 31% is flagged
   - B. 31% of all transactions are fraud
   - C. Of all flagged transactions, about 31% are really fraud
   - D. The system is 31% fast
4. Are P(A | B) and P(B | A) always equal?
   - A. No — mixing them up is a common mistake
   - B. Only when A is rare
   - C. Yes
   - D. Only for dice
5. What does the law of large numbers say?
   - A. With many trials, the observed share gets close to the true probability
   - B. Random numbers stop being random
   - C. Probabilities grow over time
   - D. Large numbers are always correct
6. A ticket costs 1,000 RWF with a 1/500 chance to win 200,000 RWF. What is the expected winning?
   - A. 200,000 RWF
   - B. 1,000 RWF
   - C. 0 RWF
   - D. 400 RWF

© WAMUGII TECH SOLUTIONS · wamugii.kento.rw · Original course material — for enrolled students.